Day 087 | Unit 06 | Teacher plan

Understand and Use Ratios and Rates

Convert Customary Units with Ratios

Standards: 6.RP.A.3d

Learning target: I can convert measurements using equivalent ratios.

enVision Topic 5: Understand and Use Ratios and Rates | Spectrum companion Chapter 3

0–5 minutes / Launch

Read the target. Ask for an estimate or a sketch before calculating. Connect the new idea to the previous lesson.

5–17 / Model

Use Worked Example 1, then Example 2. Ask students to name the next step and explain a choice. Write an equality such as 1 foot = 12 inches. Multiply to change larger units into smaller units; divide for the reverse. A conversion factor is a ratio equal to 1, so the amount stays the same.

17–42 / Practice

Assign questions 1–10 in order. Use 1–3 for foundations, 4–7 for application, and 8–10 for state-format response and reasoning. The example button provides a separate question with a worked method.

42–50 / Discuss and check

Compare representations. Review a selected answer and a plausible error. Use question 10 as an exit discussion; ask students to check units, scale, or substitution.

Support and extension

Offer number lines, fraction strips, grid paper, or a table template as appropriate. Read multi-part directions one part at a time. For extension, ask for another representation or a counterexample.

Companion and state alignment

Use the matching skill in the edition you own. No unverified publisher page numbers are assigned. AASA references below identify related public sample items by standard and format; all course questions are original.

AASA sample item references: 2

Day 087 | Unit 06 | Lesson

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01 / Learn

Write an equality such as 1 foot = 12 inches. Multiply to change larger units into smaller units; divide for the reverse. A conversion factor is a ratio equal to 1, so the amount stays the same.

Worked example 1

Convert 4 yards to feet. Show the conversion ratio.

1. Write an equality such as 1 foot = 12 inches.

2. Multiply to change larger units into smaller units; divide for the reverse.

3. 4 × 3 = 12.

4. Smaller units require more units for the same amount.

5. State the result in the form requested: 12 feet.

Answer: 12 feet

Worked example 2

Use 1 feet = 12 inches. Select all statements equivalent to 9/2 feet.
A. 54 inches
B. 3/8 inches
C. 55 inches
D. 9/2 × 12 inches

1. Write an equality such as 1 foot = 12 inches.

2. There are 12 smaller units in each larger one.

3. Multiply 9/2 by 12, giving 54 inches.

4. The physical length does not change.

Answer: A, D - 54 inches; 9/2 × 12 inches

Day 087 | Your turn

Work in order: foundation → application → explanation. Show calculations and label units.

01 / Constructed response

Convert 6 yards to feet. Show the conversion ratio.

02 / Constructed response

Convert 6 feet to inches. Show the conversion ratio.

03 / Constructed response

Convert 5 yards to feet. Show the conversion ratio.

04 / Constructed response

Convert 84 inches to feet. Show the conversion ratio.

05 / Constructed response

Convert 3 feet to inches. Show the conversion ratio.

06 / Constructed response

Convert 36 inches to feet. Show the conversion ratio.

07 / Constructed response

Convert 3/2 feet to inches. Show the conversion ratio and explain why the numerical value grows.

08 / Multiple select

Use 1 feet = 12 inches. Select all statements equivalent to 7/2 feet.
A. 42 inches
B. 43 inches
C. 7/2 × 12 inches
D. 7/24 inches

09 / Multiple select

Use 1 feet = 12 inches. Select all statements equivalent to 13/2 feet.
A. 78 inches
B. 79 inches
C. 13/2 × 12 inches
D. 13/24 inches

10 / Constructed response

Convert 6 feet to yards. Show the conversion ratio.

Day 087 | Unit 06 | Answer key

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1. 18 feet

6 × 3 = 18. Smaller units require more units for the same amount.

6.RP.A.3d | Constructed response

2. 72 inches

6 × 12 = 72. Smaller units require more units for the same amount.

6.RP.A.3d | Constructed response

3. 15 feet

5 × 3 = 15. Smaller units require more units for the same amount.

6.RP.A.3d | Constructed response

4. 7 feet

84 ÷ 12 = 7; 1 feet = 12 inches.

6.RP.A.3d | Constructed response

5. 36 inches

3 × 12 = 36. Smaller units require more units for the same amount.

6.RP.A.3d | Constructed response

6. 3 feet

36 ÷ 12 = 3; 1 feet = 12 inches.

6.RP.A.3d | Constructed response

7. 18 inches

3/2 × 12 = 18. Smaller units require more units for the same amount.

6.RP.A.3d | Constructed response

8. A, C - 42 inches; 7/2 × 12 inches

There are 12 smaller units in each larger one. Multiply 7/2 by 12, giving 42 inches. The physical length does not change.

6.RP.A.3d | Multiple select

9. A, C - 78 inches; 13/2 × 12 inches

There are 12 smaller units in each larger one. Multiply 13/2 by 12, giving 78 inches. The physical length does not change.

6.RP.A.3d | Multiple select

10. 2 yards

6 ÷ 3 = 2; 1 yards = 3 feet.

6.RP.A.3d | Constructed response